

Exploring Quantum Computing in Particle Physics

Alex McCarron

Goal:

Determine whether **Hamiltonian Truncation Effective Theory**
can **reduce digitisation errors** and/or **qubit requirements**
compared to basic **JLP** and **HO basis methods** in 0+1D scalar
field theory

Quantum Computing

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Why?

- Fundamental computational limits
- Many body quantum systems
- Emergence of NISQ Era hardware

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Qubits:

- **A quantum two-level system:** $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$
- Information stored in probability amplitudes
- For n qubits our working **Hilbert Space Dimension is $2^n = n_{\Phi}$**

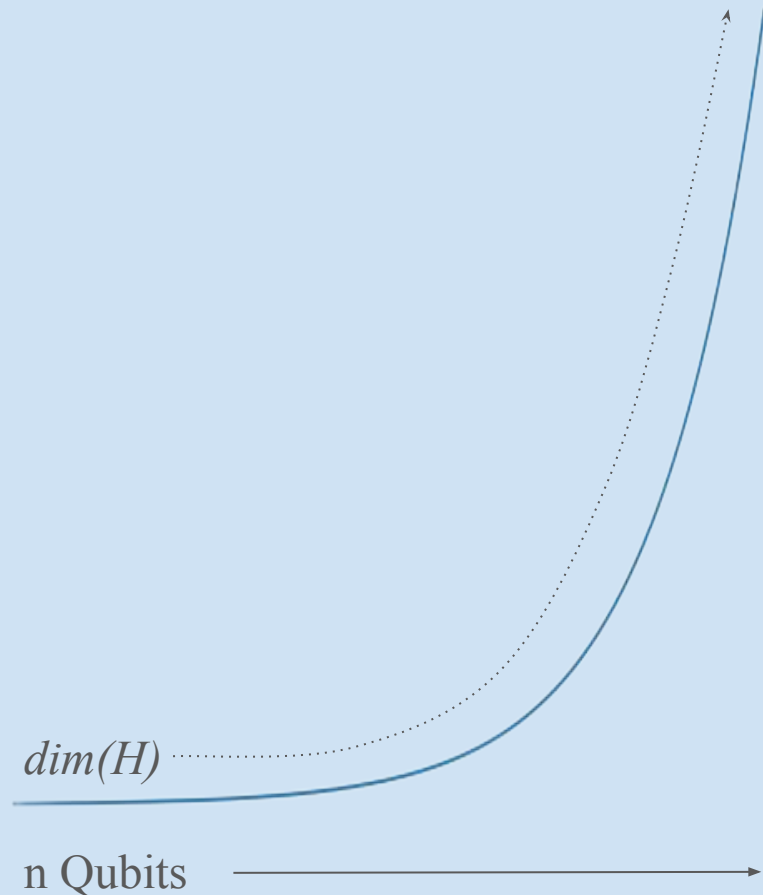
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What we're going to look at

How do we simulate Quantum Field Theories on Quantum Computers?

Digitisation Methods - What does digitisation mean? What are they?

- JLP Digitisation
- Harmonic Oscillator Digitisation
- How do they compare? How does this relate to Qubit usage?

Improving Digitisation Methods? - Can it be done?

- Hamiltonian Truncation Effective Theory (HTET)
- Does it improve errors?
- Does it decrease the number of Qubits we need?

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Truncation + Discretisation

- We set a maximum and minimum value
- We split a continuous field into finite parts

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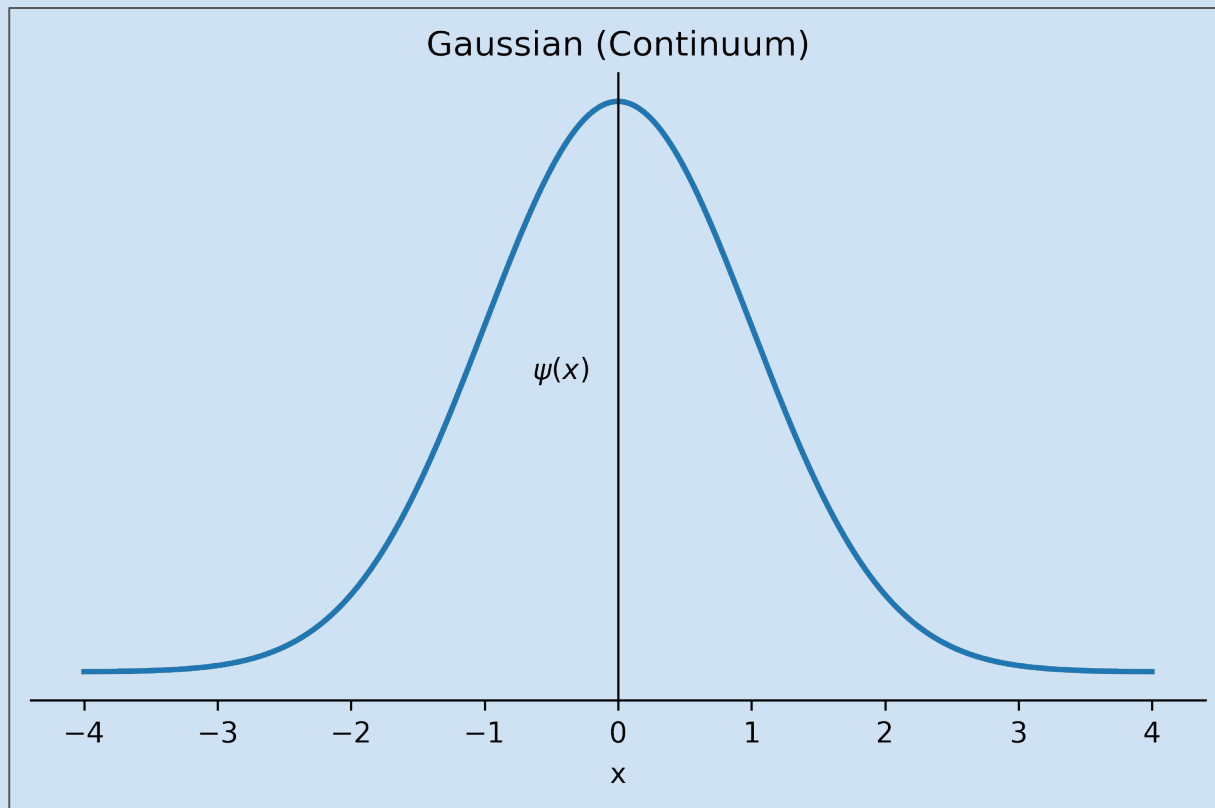
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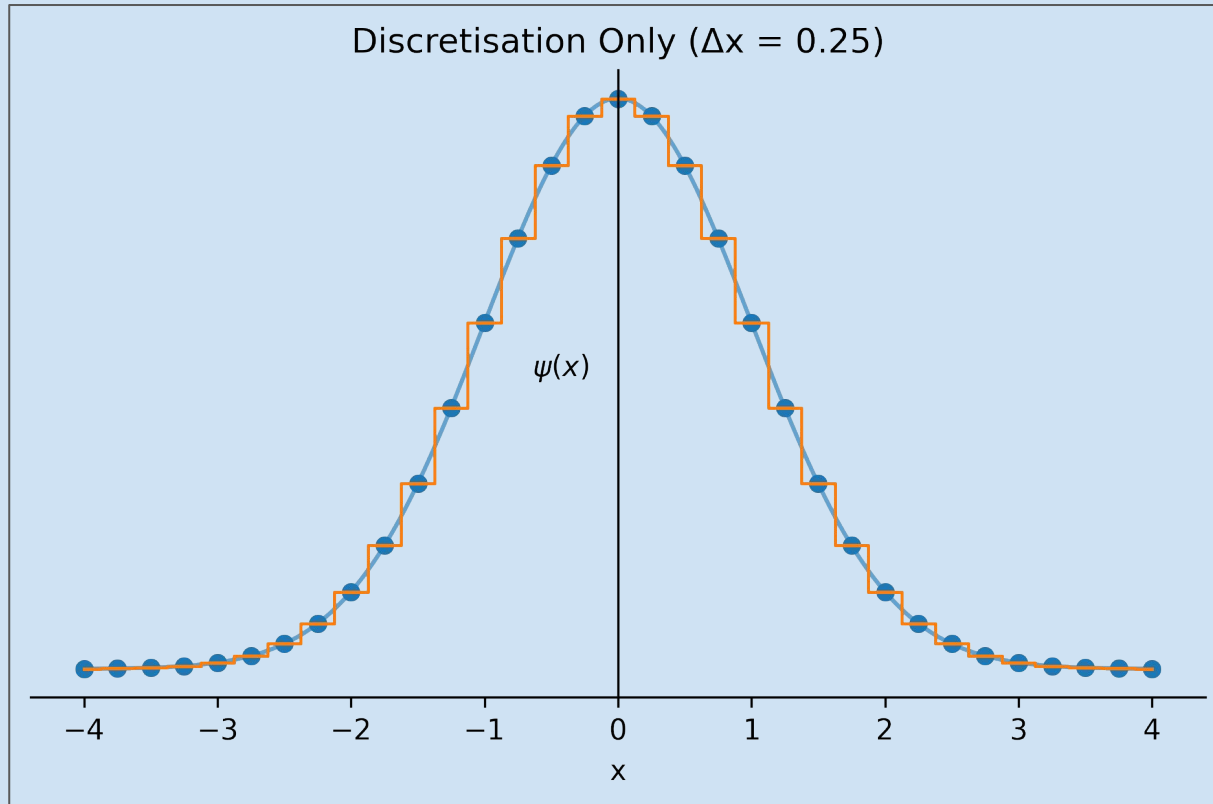
ERRORS

Digitisation - How does it work?

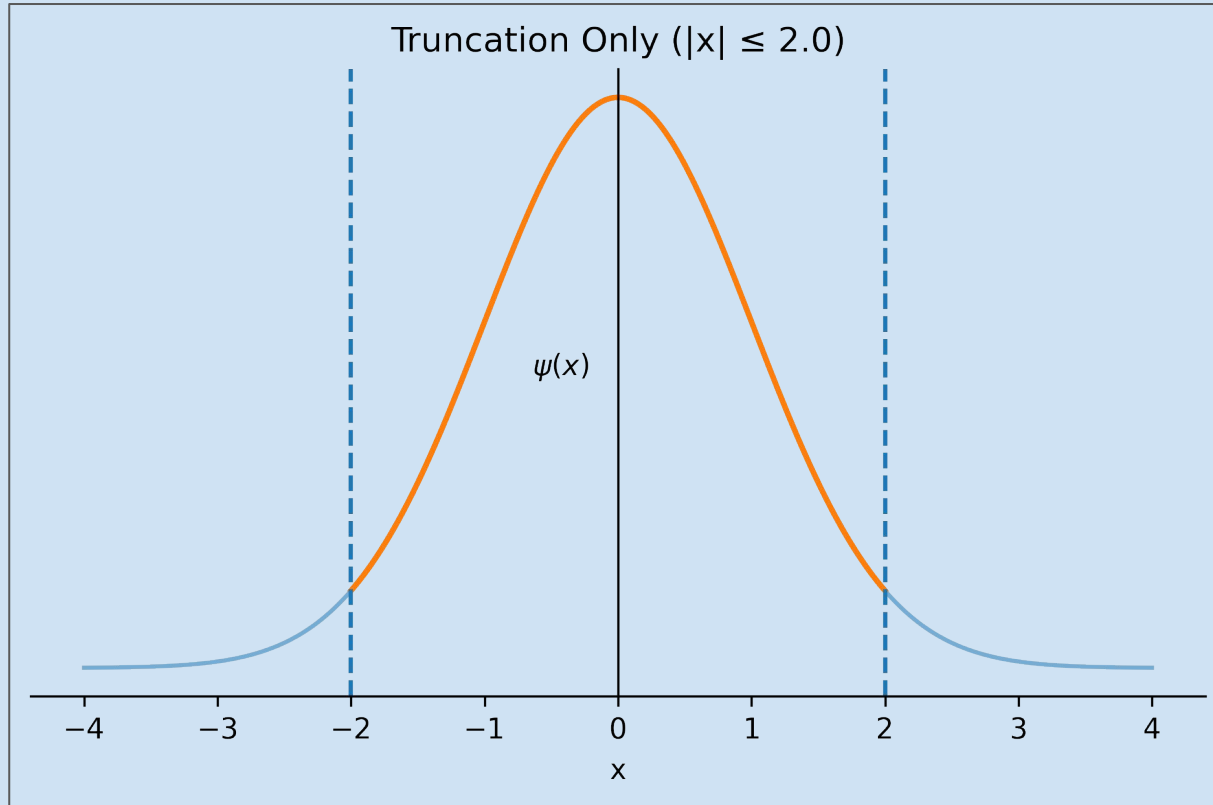
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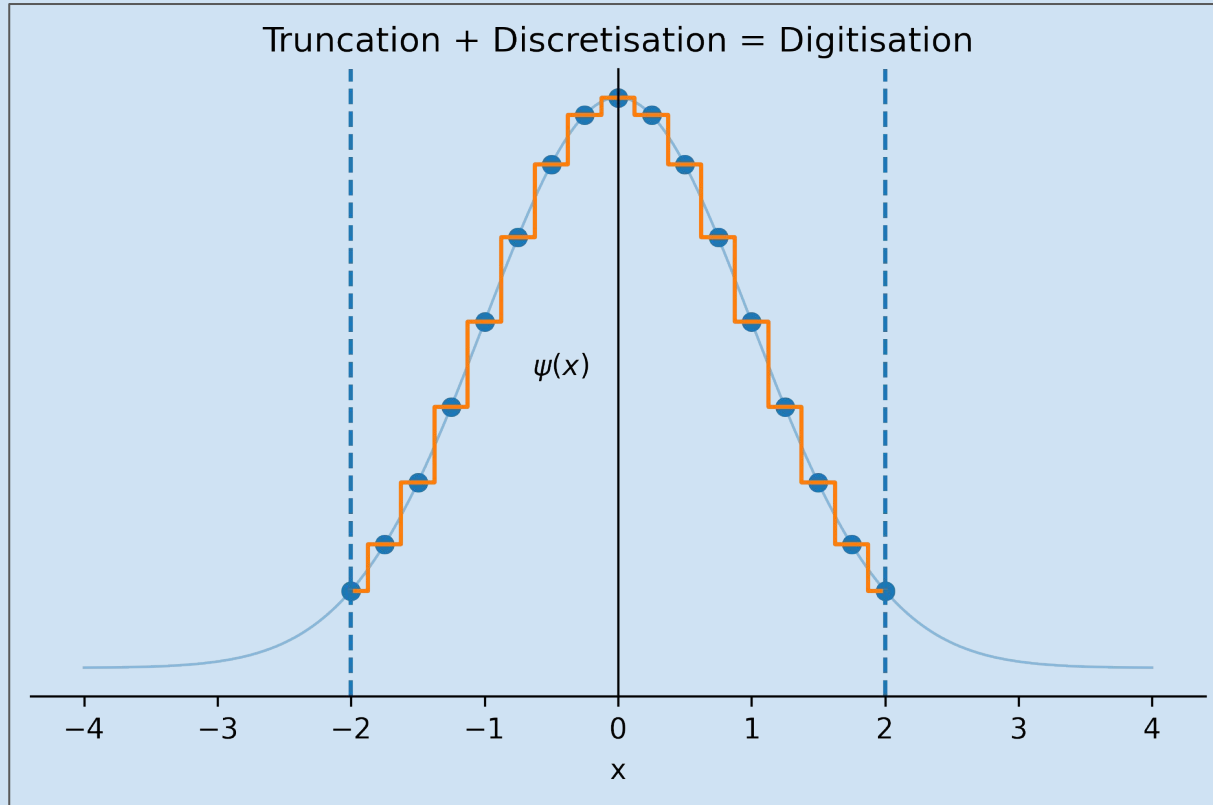
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(0+1D Quantum Mechanics)

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- Compare against known values

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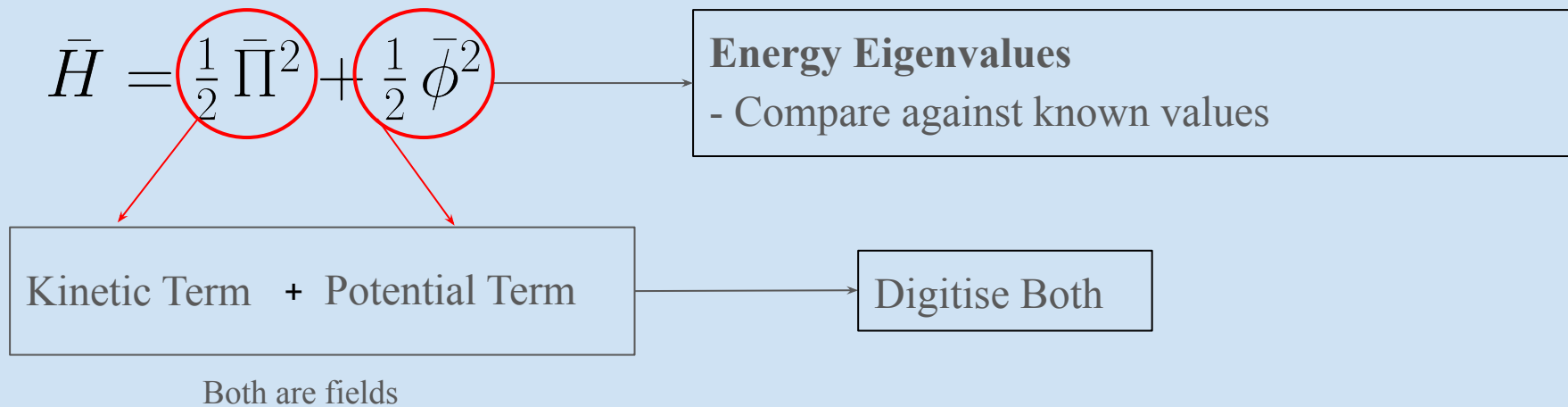
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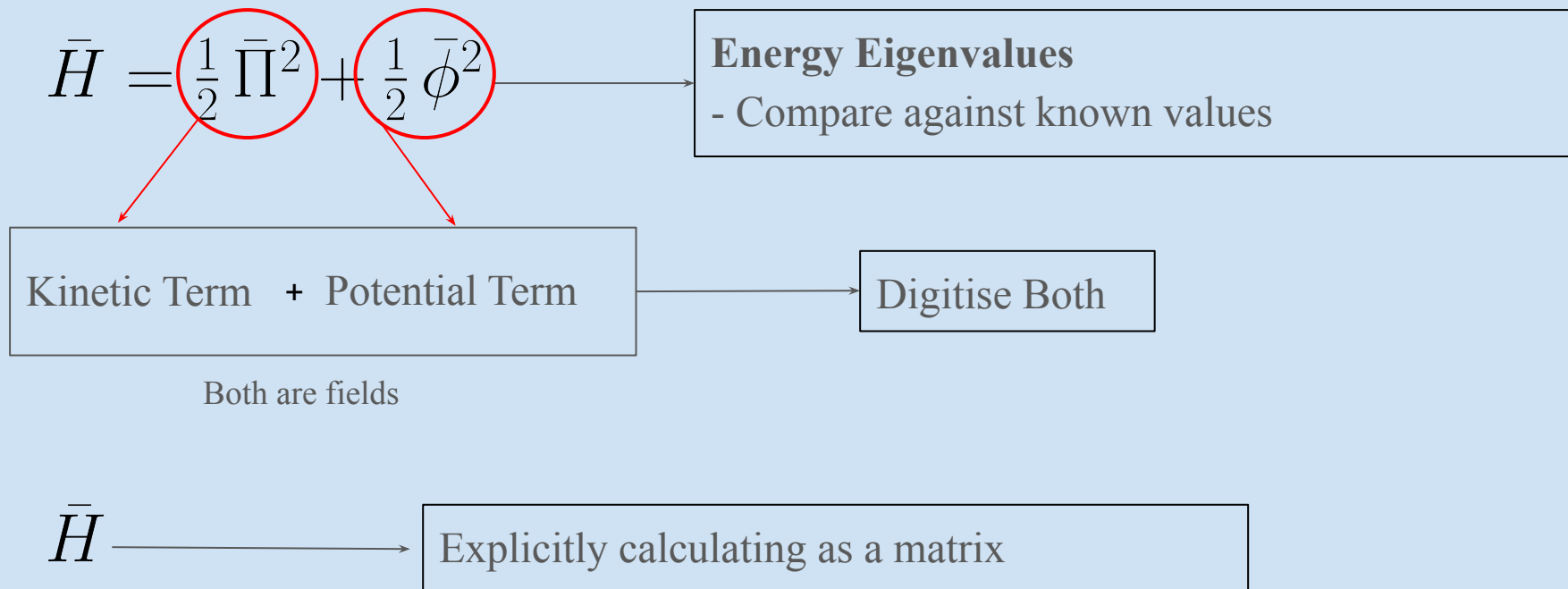
Kinetic Term + Potential Term

Both are fields

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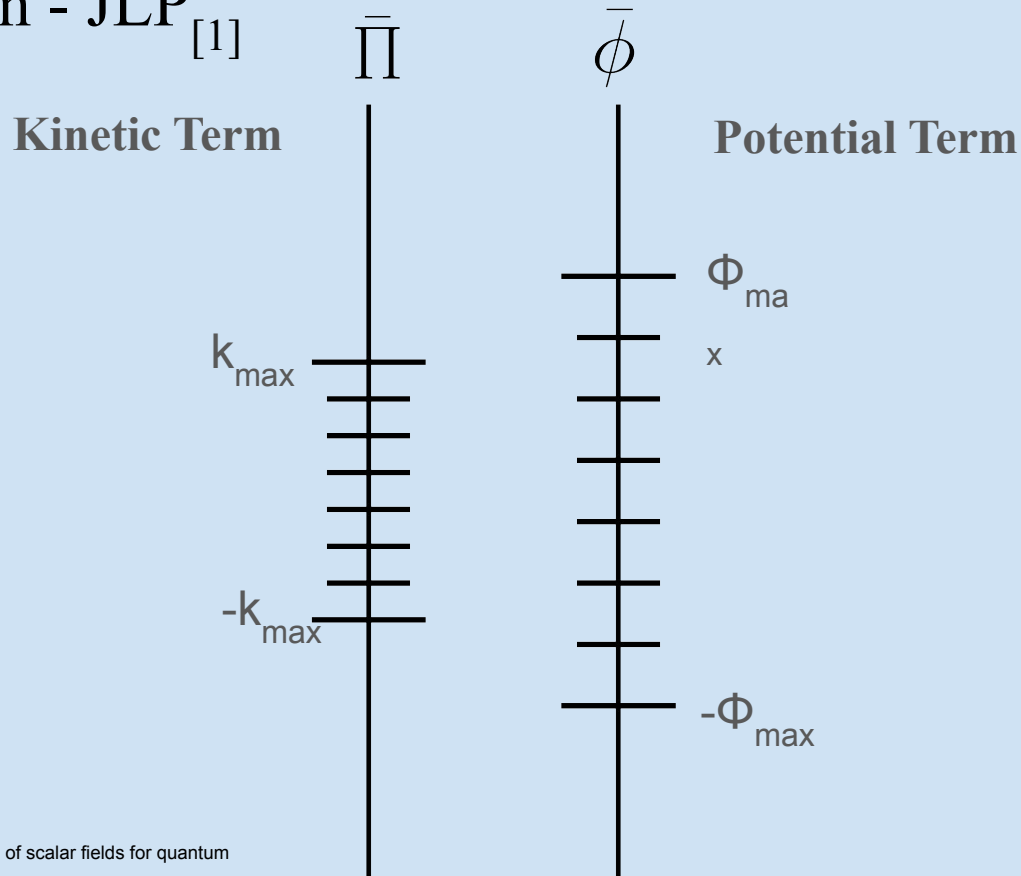
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$$\bar{\Pi}$$

$$\bar{\phi}$$

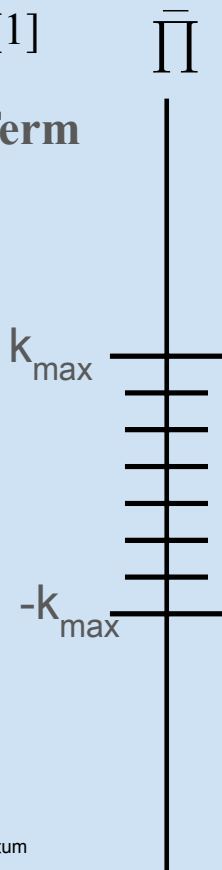
Potential Term

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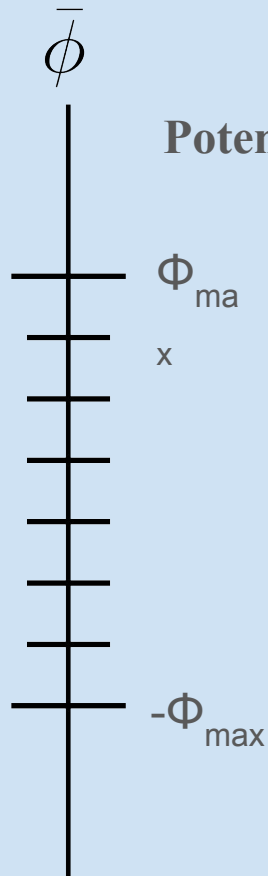


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Kinetic Term



Potential Term

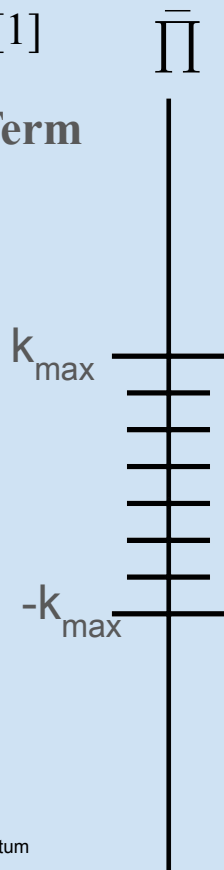


Matrix Construction

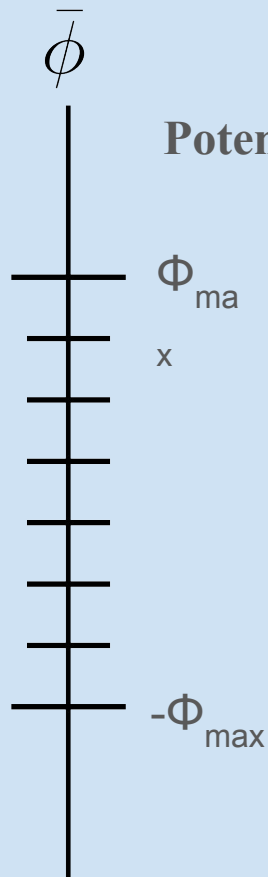
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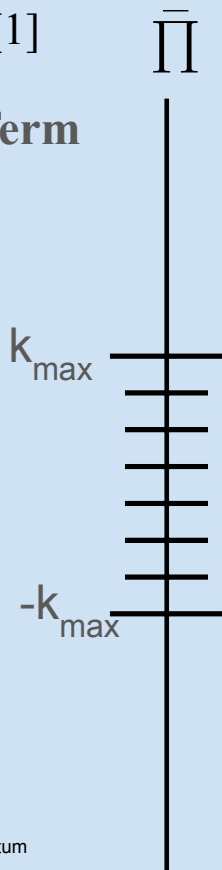
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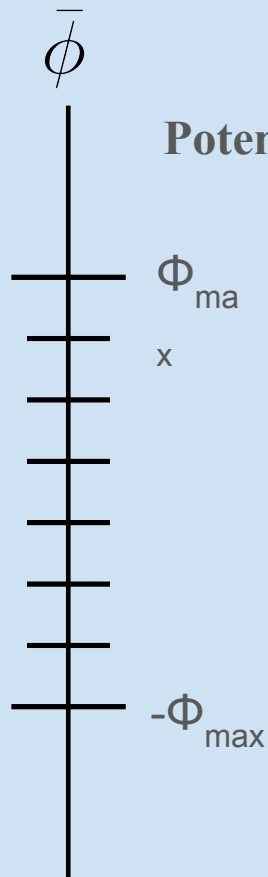
Kinetic Term

$$k_{max} = \frac{\pi}{\delta\phi}$$

$$\delta k = \frac{2k_{max}}{n_\phi}$$



Potential Term



$$\delta\phi = \frac{2\phi_{max}}{n_\phi - 1}$$

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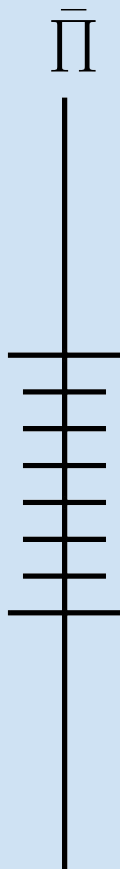
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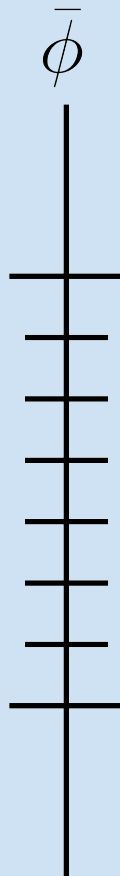
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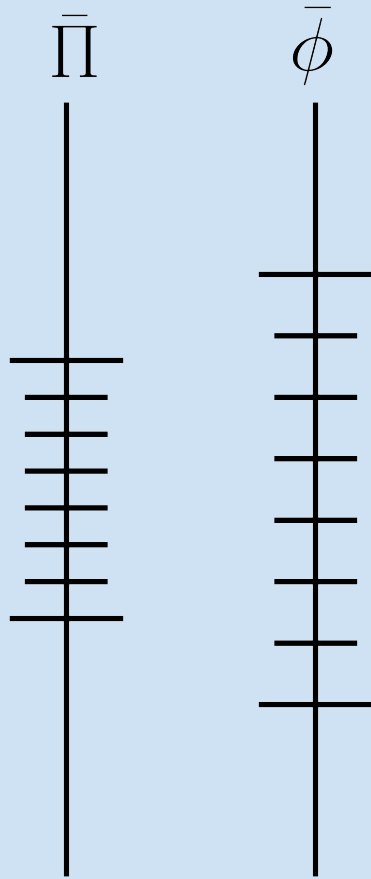


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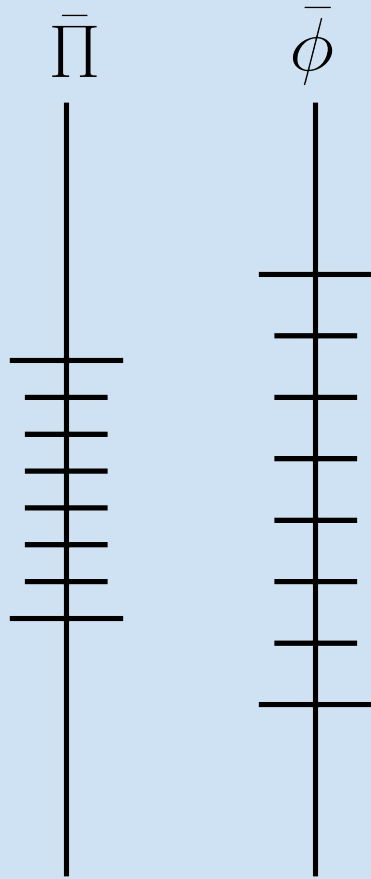
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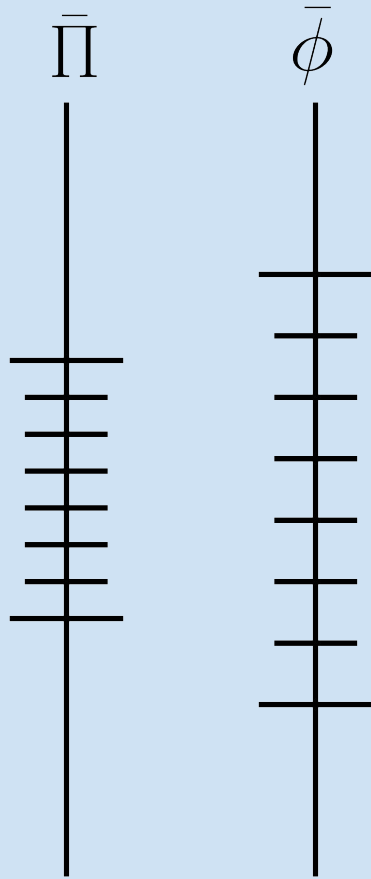


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Nyquist Shannon Sampling Theorem:

- Balance contributions from both the kinetic and potential terms

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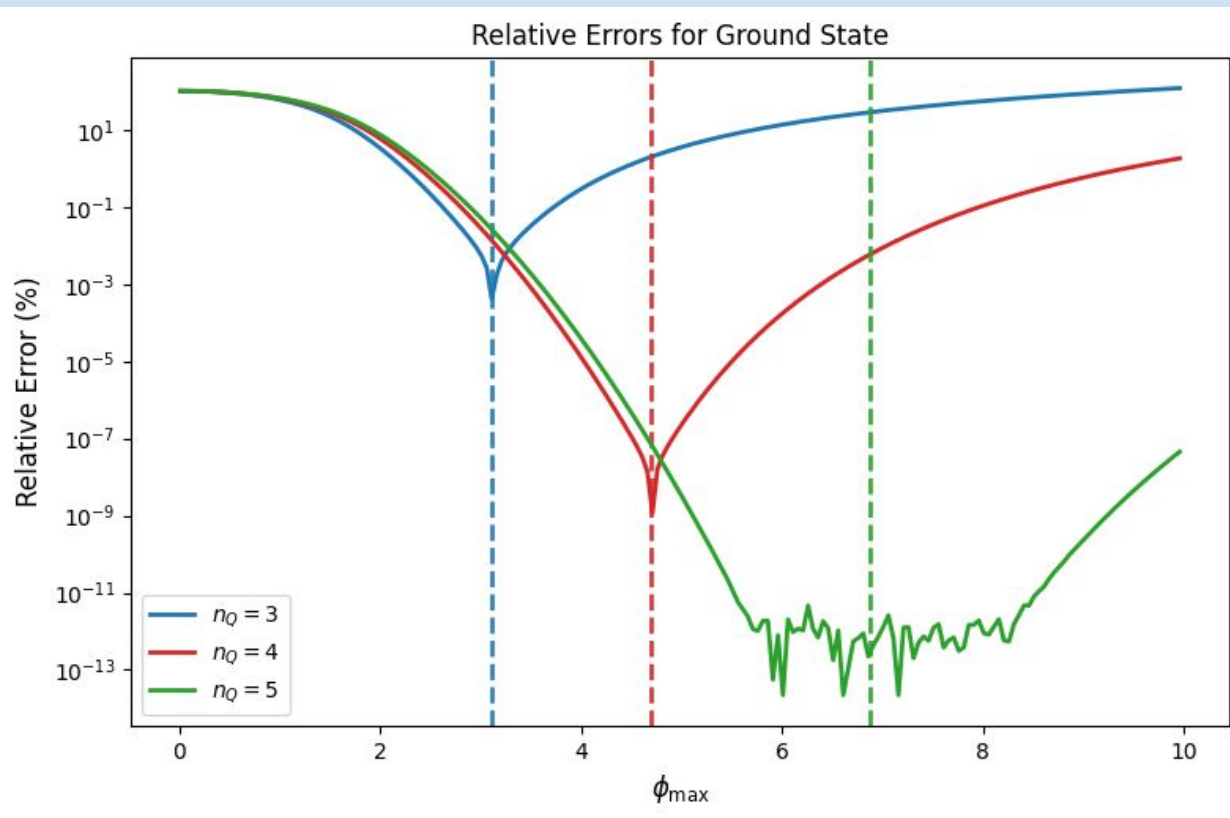
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$$\phi_{\max} = \frac{1}{\sqrt{\omega}} \sqrt{\frac{\pi(n_{\phi}-1)^2}{2n_{\phi}}} \quad \omega = 1 \quad [4]$$

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[4] - C. W. Bauer, B. Nachman, and M. Freytsis, Simulating collider physics on quantum computers using effective field theories, Physical Review Letters 127, 10.1103/physrevlett.127.212001 (2021).

Digitisation - JLP_[1]



Dotted lines are calculated optimal $\phi_{\max}$ values

Decreased errors for increased Qubit number

Direct reproduction of results seen in literature ^[1]

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Digitisation - Harmonic Oscillator Basis ^[1]

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Digitise a **set of harmonic oscillator energy eigenstates** instead of the field and momentum spaces directly

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How do we express $\bar{\Pi}$ and $\bar{\phi}$ in terms of these states?

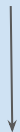
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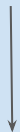
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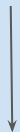
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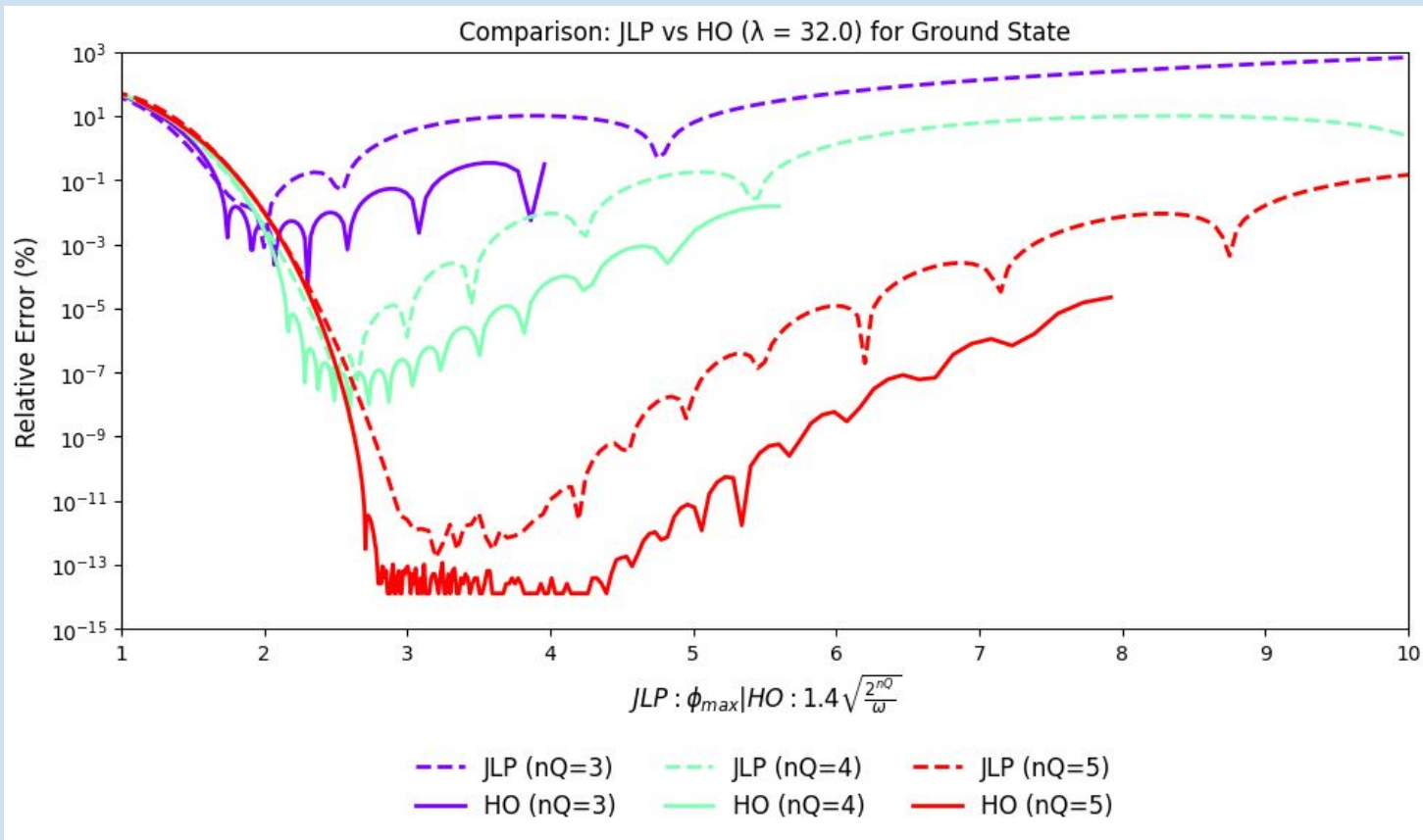
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Digitisation - Harmonic Oscillator Basis



Harmonic
Oscillator basis
significantly
outperform JLP at
a given number of
Qubits

Direct
reproduction of
structure seen in
literature [1]

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Hamiltonian Truncation Effective Theory ^[2] - Looking at improving Digitisation Methods

Can we **reduce error** further?

Can we reduce the **qubit requirement** for that error?

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$$(H_1)_{fi} = H$$

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$$(H_3)_{fi} = \sum_{\alpha, \beta \in Q} \frac{V_{f\alpha} V_{\alpha\beta} V_{\beta i}}{(E_f - E_\alpha)(E_f - E_\beta)} - \sum_{\alpha \in P} \sum_{\beta \in Q} \frac{V_{f\alpha} V_{\alpha\beta} V_{\beta i}}{(E_\alpha - E_\beta)(E_f - E_\beta)}$$

P is the region below $E_{\max}$, Q is the region above $E_{\max}$

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Significantly H_{eff} is **Non Hermitian**

This is justified for several reason in [2]:

- The underlying full Hamiltonian is Hermitian
- At physical eigenvalues the effective Hamiltonian reproduces the exact spectrum of the Hermitian Hamiltonian
- Asymmetry that causes Non Hermiticity vanishes as E_{max} tends to infinity

P is the region below E_{max} , Q is the region above E_{max}

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- At finite E_{max} results depend on basis choice, by varying ω we can optimise convergence and minimise truncation error (this is the same as the ω in HO basis)
- Allows us to define $H_0 + V_{\text{residual}}$
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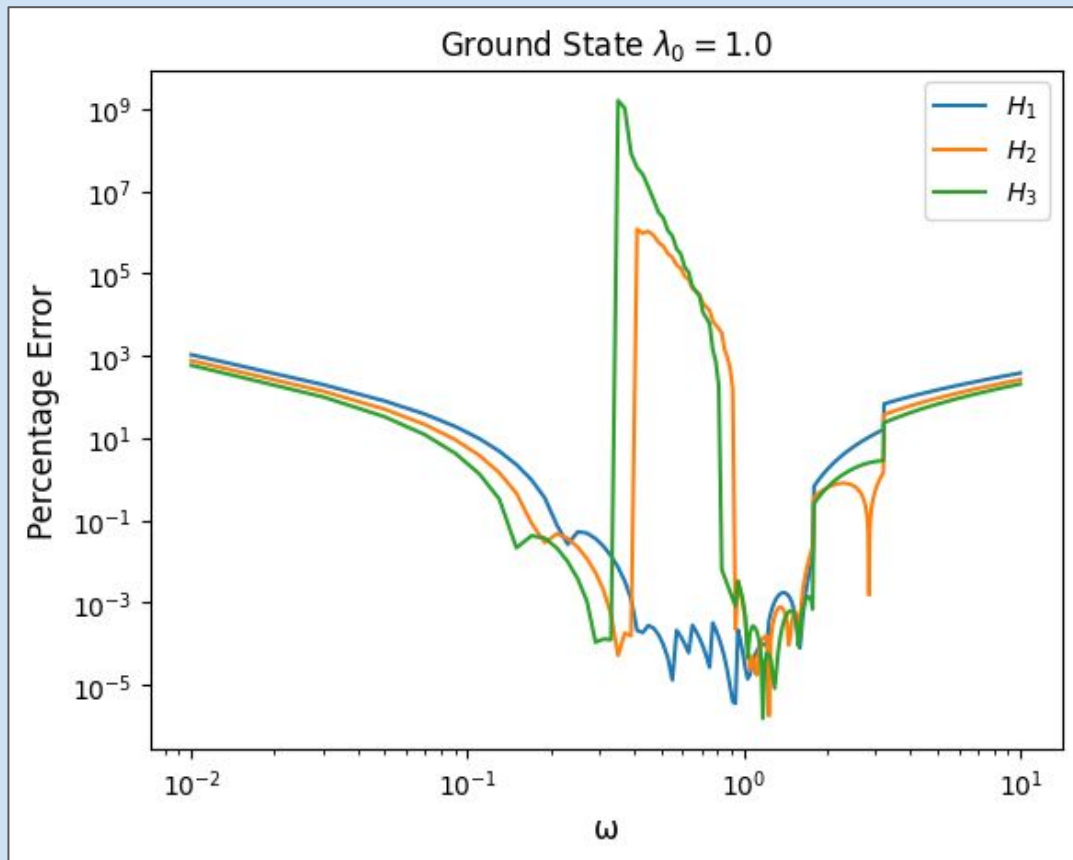
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Potential (V)

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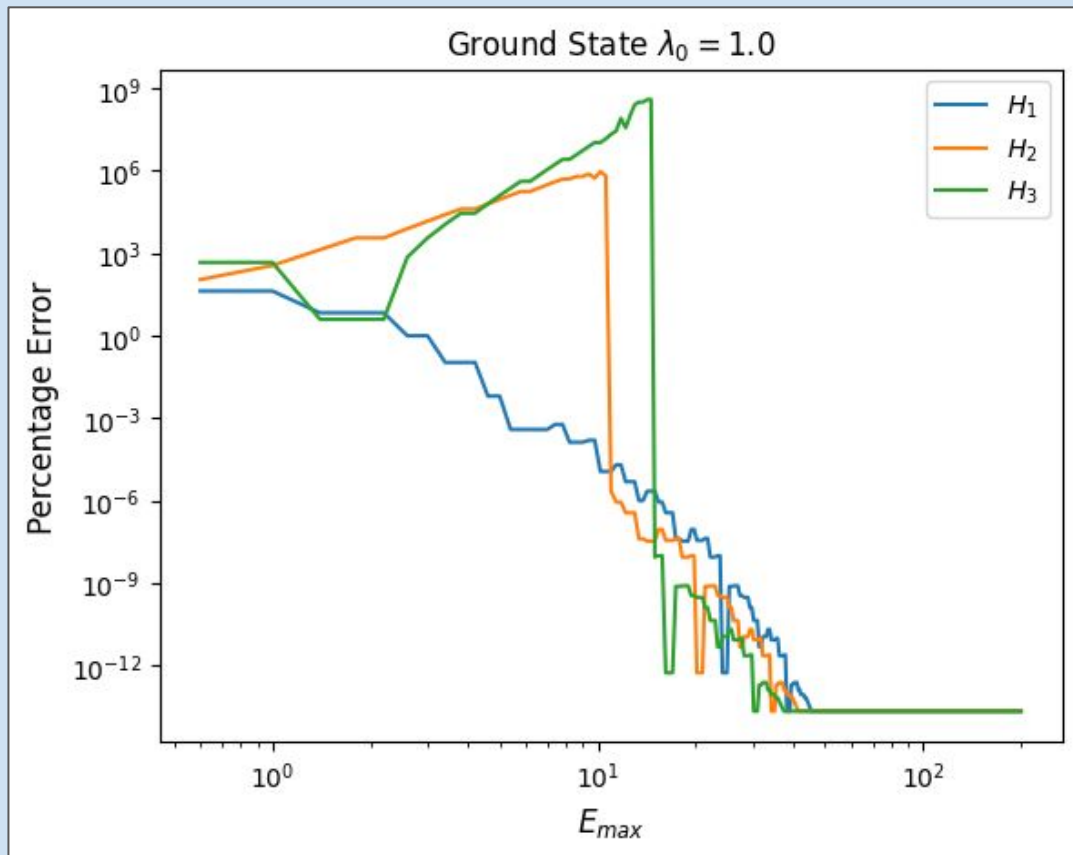
Hamiltonian Truncation Effective Theory



- Large errors at a critical $E_{\max}$ as denominators become small
- H1 mimics structure seen in [1]

$$(H_2)_{fi} = \sum_{\alpha \in Q} \frac{V_{f\alpha} V_{\alpha i}}{E_f - E_\alpha}$$

Hamiltonian Truncation Effective Theory

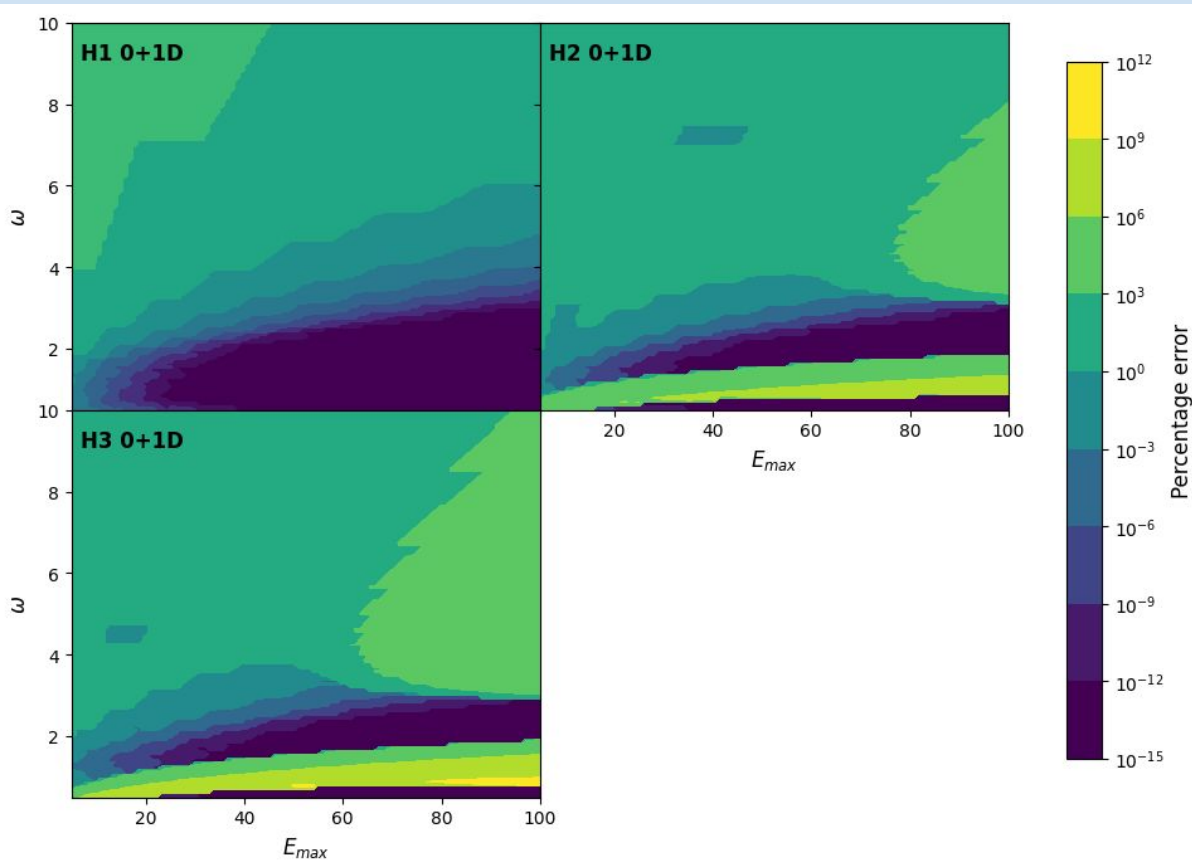


- Large errors at small E_{max} as denominators become small as $E_Q \rightarrow E_f$
- H3 achieves a minimum error at a lower number of qubits than H2 than H1
- Agreement with literature^[3] that HTET is viable for NISQ era devices

	E_{max} at minimum error
H1	38.16
H2	34.17
H3	30.17

$$\sum_{\alpha \in Q} \frac{V_{f\alpha} V_{\alpha i}}{E_f - E_\alpha}$$

Hamiltonian Truncation Effective Theory



General expected structure from previous two graphs

H3 achieves a minimum error at a lower number of qubits than H2 than H1

Lowest values of error depend on choice of ω

	E_{max} at minimum error
H1	37.31
H2	33.42
H3	29.86

Goal: Determine whether **Hamiltonian Truncation Effective Theory** can **reduce digitisation errors** and/or **qubit requirements** compared to basic **JLP** and **HO basis methods** in 0+1D scalar field theory

We looked at:

- What digitisation is
- Methods of digitisation
- Which of JLP or HO basis had reduced errors
- What Hamiltonian Truncation Effective Theory is
- Can it be effective in reducing the errors of digitisation

HTET can reduce digitisation and/or qubit requirements within 0+1D compared to digitisation methods **provided you are able to correctly tune ω**